IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-150

1. Topic (12 words max)

AI-Powered Tour Guide Booking System with Voice and RAG

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information System Engineering(ISE)

4. If a continuation of a previous project:

Project ID	25-26J-150
Year	2025

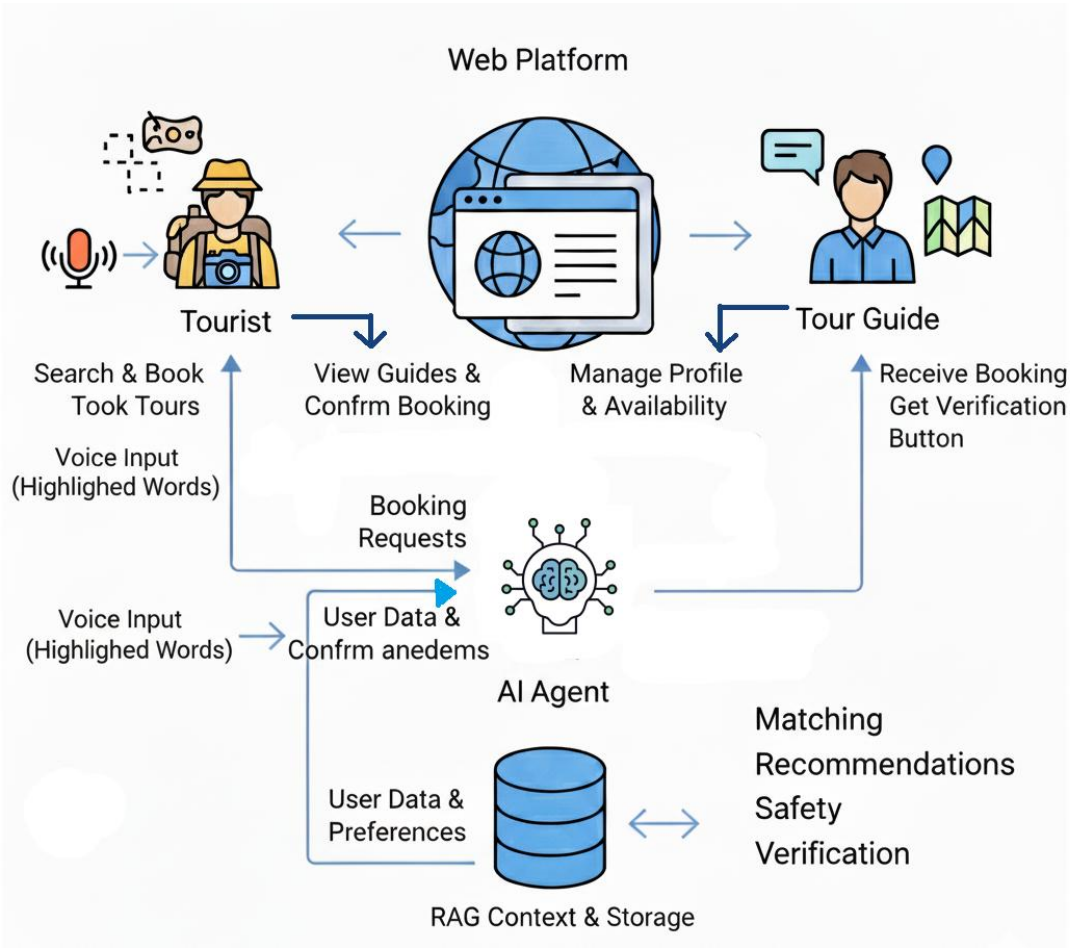
5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Tourism is a vital sector in Sri Lanka's economy, attracting millions of visitors annually due to its cultural heritage, biodiversity, and natural attractions. However, one of the persistent challenges faced by tourists is the difficulty of finding trusted, verified, and suitable tour guides who align with their preferences, language needs, and budget. Existing platforms such as TripAdvisor, GetYourGuide, and Viator provide basic booking and review mechanisms, but they often lack robust multi-layered verification, resulting in risks of fraud, safety issues, and mismatched services. Tourists may encounter unverified guides, misleading reviews, or premature exposure of personal information, leading to reduced trust in online guide-booking systems.

Furthermore, current platforms fail to leverage AI-driven personalization for effective tourist–guide matching. The recommendation systems used are mostly static, relying on ratings or broad categories, without incorporating dynamic user preferences, cultural contexts, or real-time data such as weather, festivals, and safety advisories. This gap results in suboptimal travel experiences where tourists struggle to obtain relevant and timely guidance. In addition, the absence of advanced fraud detection mechanisms and privacy-preserving workflows exposes tourists and guides to risks, particularly in peer-to-peer platforms where trust is a critical factor.

Given these limitations, there is a need for a secure, AI-assisted tourism platform that not only ensures the authenticity of guides through multi-document verification but also integrates Retrieval-Augmented Generation (RAG) to provide accurate, context-aware responses to tourist queries. By combining knowledge-driven recommendations, fraud detection, and privacy-protected data sharing, the system can improve both the reliability and personalization of tourism services. Such a solution would significantly enhance user trust and satisfaction while addressing the broader challenges of digital tourism, security, and intelligent service matching.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)



The solution is a web platform for on-demand tour guide bookings, similar to Uber. It features both verified and non-verified guides, ensures user privacy until a booking is confirmed, and includes crucial emergency and safety features. An integrated **AI agent** handles intelligent guide-tourist matching, fraud detection, personalized recommendations, and provides safety guidance, creating a secure and efficient experience for both tourists and guides.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The proposed research requires interdisciplinary expertise spanning web development, artificial intelligence, and cybersecurity. Proficiency in **modern web frameworks** such as React.js and Node.js/Express is essential for building a scalable and interactive booking platform, while **UX/UI design** principles ensure intuitive interfaces that protect user privacy. Knowledge of **tourism regulations** and **emergency geolocation systems** is critical to align the system with industry standards and safety requirements.

Specialized knowledge in **authentication protocols (OAuth), database encryption, and GDPR-compliant security practices** underpins the trustworthiness of the platform. Expertise in **machine learning** is required to implement guidetourist matching and personalized recommendations, while **natural language processing (NLP)** supports multilingual chatbot interactions. Additionally, **AI-assisted fraud detection** and integration with verification services (e.g., police check APIs, identity validation) strengthen the reliability of bookings.

The project also demands diverse datasets, including **user profiles, booking histories, geospatial data, and verified guide credentials**, which must be securely stored and anonymized for training. Integration with **external verification APIs** and **social media links** provides richer contextual data. Collectively, these requirements ensure the system delivers a secure, intelligent, and privacy-preserving solution for personalized tour guide bookings.

8. Objectives and Novelty

Main Objective Develop a secure web application for tour guide bookings that incorporates multi-level verification, privacy-protected detail unlocking, AI-assisted matching and recommendations, improving trust and efficiency in tourism services.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
P.R.I.PRAVEAN IT21191060	1. Design user interfaces for tourist voice/text queries, guide registration with document uploads, and booking inputs. 2. Implement a tiered categorization system for verified and non-verified guides with multi-document verification. 3. Develop AI-based matching and recommendation engine using RAG, considering user preferences, highlighted keywords,	1. Requirement gathering and competitor analysis to identify gaps in existing tourism booking systems. 2. System architecture design (frontend, backend, vector database schema, encrypted data storage, APIs for voice input and verification). 3. Build 5 core modules are Document verification workflows, Voice	1. Voice-enabled tourism query system that extracts highlights keywords and delivers contextual responses. 2. Tiered verification model with multi-document checks (government ID, license, police verification, optional social media). 3. Privacy-aware booking flow where sensitive details are unlocked only after confirmation.

	verification status, and historical booking data. 4. Integrate privacy and safety mechanisms, including personal detail locking/unlocking post-booking, emergency location tracking, and feedback systems. 5. Embed AI agent features for multilingual chatbot support, personalized itinerary suggestions, fraud detection, and scam alerts. 6. Ensure platform security, scalability, and performance through encryption, cross-browser testing, and backend optimization.	input processing & keyword extraction, AI agent (chatbot + RAG retrieval), Privacy locking and unlocking features, Emergency response and feedback system. 4. Conduct security audits, usability testing, and multilingual validation (Sinhala, Tamil, English). 5. Deploy and evaluate the system through beta testing with key metrics: booking success rate, AI response accuracy, fraud detection rate, and user satisfaction.	4. AI agent with RAG integration for personalized suggestions, intelligent guide matching, fraud detection, and real-time safety guidance. 5. Balanced trust model combining accessibility, privacy, multilingual AI support, and scam prevention — not available in current competitor systems.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
P.R.I.PRAVEAN IT21191060	Illustrates design for voice interfaces and intelligent structured data modeling, knowledge base construction, agents. Emphasize and voice interaction patterns. It ensures software maintainability and curriculum-focused knowledge delivery through structured system workflows.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms	Suranjini	Silva	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.